OBS Studio Current Settings Guide

NHLC profile and NHLC Axis Audio scene collection

Revised 2026-06-27 - section numbering corrected; OBS backup and global restore section added.

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1. Scope and source files reviewed

Source	File / location	Used for
OBS profile folder	basic/profiles/NHLC	Identifies the profile name and stores video, audio, output, encoder, recording, and some saved YouTube-interface settings.
Scene collection	basic/scenes/NHLC_Axis_Audio.json	Defines the active OBS scene collection: scenes, sources, source properties, filters, source mute state, and scene item placement.
Encoder files	streamEncoder.json and recordEncoder.json	Confirm the current H.264 hardware-encoder bitrate and keyframe interval values.
OBS WebSocket config	plugin_config/obs-websocket/config.json	Confirms that OBS WebSocket is enabled on port 4455 and authentication is currently off.
Global OBS config	global.ini	Provides global OBS behavior such as Direct3D renderer, browser hardware acceleration, and update setting.
Latest OBS log in	logs/2026-06-27 09-44-26.txt	Confirms OBS version, PC environment, plugin load

uploaded backup		status, DistroAV/NDI, obs-websocket, source initialization, and media-source details.
Redacted stream service file	service.json	Confirms the custom YouTube RTMPS server with stream key intentionally replaced by placeholder text.

2. High-level OBS design

The current OBS design is a simple three-scene Axis-camera layout intended to be driven by Stream Agent III sd. There are two live camera scenes and one introduction scene. The two live scenes use Axis camera RTSP Media Sources that carry both video and embedded camera audio. Proclaim overlay graphics arrive as an NDI source named slides and are placed over the camera scenes.

Item	Current setting / purpose
Profile	NHLC
Scene collection	NHLC Axis Audio
Scene names that matter to Stream Agent	Introduction, West View, East View
Camera sources	West_axis and East_axis
Primary current audio path	Axis embedded audio in the West_axis and East_axis RTSP Media Sources
Proclaim overlay path	DistroAV / NDI source named slides: BEELINKOBS (Proclaim - proclaim overlays)
Intro source	Intro_Video media source: C:/ChurchAutomation/Intro Video/20_second_greeting.mp4

Origin	Transport	OBS source	OBS scene	Result
West camera	RTSP TCP with embedded audio	West_axis source	West View scene	OBS stream/record
East camera	RTSP TCP with embedded audio	East_axis source	East View scene	OBS stream/record
Proclaim	NDI transparent overlay	slides source	West/East View scenes	Visual overlay only; source is muted
Intro video file	Local media file	Intro_Video source	Introduction scene	Played by Stream Agent at stream start

3. OBS profile settings - NHLC

Setting	Current value / note
OBS version seen in latest log	OBS 32.0.4 (64-bit, windows)
Computer seen in latest log	AMD Ryzen 7 6800H with Radeon Graphics; 28451MB Total, 21031MB Free
Output mode	Advanced
Video canvas / output	1920 x 1080 base; 1920 x 1080 output
Frame rate	30 fps
Downscale filter	bicubic
Color format / space / range	NV12 / Rec. 709 / Partial
Audio sample rate / channel setup	48000 Hz / Stereo
Audio monitoring device	Default (default)
Auto remux	true

Streaming output setting	Current value
Streaming encoder	h264_texture_amf (AMD HW H.264 / AVC according to log)
Streaming bitrate	8000 kbps
Streaming keyframe interval	2 seconds
Streaming audio encoder	ffmpeg_aac
Streaming audio track	Track 1
Track 1 bitrate	256 kbps
Apply service settings	true
Reconnect enabled	true - retry delay 2 seconds, max retries 25
Low latency network option	true

Recording output setting	Current value
Recording type	Standard
Recording folder	C:\\Stream Recordings
Recording format	mp4
Recording encoder	h264_texture_amf (AMD HW H.264 / AVC)
Recording bitrate	8000 kbps
Recording keyframe interval	2 seconds
Recording audio encoder	ffmpeg_aac
Recording audio tracks	1
Split recording files	Type=Time; time=15 minutes; size=2048 MB

Note: The profile currently records directly to MP4. MP4 is convenient but less crash-safe than MKV during a power failure or OBS crash. If future reliability testing favors MKV, change the recording format and keep Auto Remux enabled.

4. Streaming service and stream-key handling

Service setting	Current value / documentation rule
Service type	rtmp_custom
Server	rtmps://a.rtmps.youtube.com/live2
Use authentication	False
Bandwidth test mode	False
Stream key	Redacted in uploaded file. Do not place the real key in GitHub or printed documentation.

WARNING: The uploaded basic.ini also contains saved YouTube integration values, including old OAuth-style token fields and channel metadata. Those values should be treated as sensitive and possibly stale. The production stream key should live only in OBS on the streaming PC or in a carefully protected private backup.

5. Scene collection - NHLC Axis Audio

Setting	Current value / note
Scene collection name	NHLC Axis Audio
Scene order	Introduction, West View, East View
Current scene in backup	East View

Current program scene in backup	East View
Current transition in backup	Fade at 350 ms
Quick transitions saved	Cut 300 ms; Fade 300 ms; Fade 300 ms fade-to-black

Note: Stream Agent III sd depends on exact scene and source names. Changing Introduction, West View, East View, West_axis, East_axis, slides, or Intro_Video requires matching Stream Agent configuration changes.

Scene	Draw order	Source item	Visible	Position	Scale	Crop
West View	1	West_axis	True	x=0.0, y=0.0	x=1.0, y=1.0	None
West View	2	slides	True	x=0.0, y=0.0	x=1.0, y=1.0	None
Introduction	1	West_axis	True	x=1.0, y=2.0	x=1.0, y=1.0	None
Introduction	2	Intro_Video	True	x=0.0, y=0.0	x=1.0, y=1.0	None
East View	1	East_axis	True	x=0.0, y=0.0	x=1.0, y=1.0	None
East View	2	slides	True	x=0.0, y=0.0	x=1.0, y=1.0	None

Note: The Introduction scene has West_axis as a camera/background source and Intro_Video as the full-frame greeting video. West View and East View each include their camera source and the Proclaim slides overlay source.

6. Source-by-source settings

6.1 West_axis camera source

Setting	Current value / note
OBS source name	West_axis
OBS source type	ffmpeg_source
Enabled / muted	enabled=True; muted=False
Volume	linear=0.7445892691612244 (-2.6 dB)
Audio sync offset	0 ms
Monitoring	Monitor off
Mixer track note	OBS mixer bitmask=193. Track 1 is selected in the profile output, and these sources include Track 1.
RTSP input	rtsp://<camera-login>@192.168.88.2:554/axis-media/media.amp?streamprofile=OBS_1080p30_audio&audio=1
Hardware decode	True
FFmpeg options	rtsp_transport=tcp rtsp_flags=prefer_tcp
Local file	False
Restart playback when source becomes active	False
Reconnect delay	1 second(s)

Filter name	Filter type	Enabled	Documented setting
VST 2.x Plug-in	vst_filter	Yes	plugin=C:/Program Files/Steinberg/VstPlugins/TDR Nova.dll; settings stored as VST chunk; summarized below
Noise Suppression	noise_suppress_filter	Yes	method=speex; suppress_level=-20 dB
3-Band Equalizer	basic_eq_filter	Yes	3-band equalizer filter present; no explicit band settings stored in JSON

Gain	gain_filter	Yes	gain=1.8 dB

TDR Nova band	Active	Gain dB	Freq Hz	Q	Type
1	On	-10.5	60	3.44	Bell
2	On	0.0	400	0.60	Bell
3	On	0.0	2200	0.50	Bell
4	On	0.0	6000	0.60	High S

Note: TDR Nova settings are listed only as a human-readable summary. The authoritative setting is the VST plug-in preset/chunk stored inside the OBS scene collection.

6.2 East_axis camera source

Setting	Current value / note
OBS source name	East_axis
OBS source type	ffmpeg_source
Enabled / muted	enabled=True; muted=False
Volume	linear=0.7445892691612244 (-2.6 dB)
Audio sync offset	0 ms
Monitoring	Monitor off
Mixer track note	OBS mixer bitmask=255. Track 1 is selected in the profile output, and these sources include Track 1.
RTSP input	rtsp://<camera-login>@192.168.88.3:554/axis-media/media.amp?streamprofile=OBS_1080p30_audio&audio=1
Hardware decode	True
FFmpeg options	rtsp_transport=tcp rtsp_flags=prefer_tcp
Local file	False
Restart playback when source becomes active	False
Reconnect delay	1 second(s)

Filter name	Filter type	Enabled	Documented setting
VST 2.x Plug-in	vst_filter	Yes	plugin=C:/Program Files/Steinberg/VstPlugins/TDR Nova.dll; settings stored as VST chunk; summarized below
Noise Suppression	noise_suppress_filter	Yes	method=speex; suppress_level=-20 dB
3-Band Equalizer	basic_eq_filter	Yes	3-band equalizer filter present; no explicit band settings stored in JSON
Gain	gain_filter	Yes	gain defaults to 0 dB in the saved JSON

TDR Nova band	Active	Gain dB	Freq Hz	Q	Type
1	On	-11.7	62	5.00	Bell
2	On	-11.3	245	5.00	Bell
3	Off	0.0	2200	0.50	Bell
4	Off	0.0	6000	0.60	High S

Note: TDR Nova settings are listed only as a human-readable summary. The authoritative setting is the VST plug-in preset/chunk stored inside the OBS scene collection.

6.3 slides - Proclaim NDI overlay source

Setting	Current value / note
OBS source name	slides
OBS source type	ndi_source
Enabled / muted	enabled=True; muted=True
Volume	linear=1.0 (0.0 dB)
Audio sync offset	0 ms
Monitoring	Monitor off
Mixer track note	OBS mixer bitmask=193. Track 1 is selected in the profile output, and these sources include Track 1.
NDI source name	BEELINKOBS (Proclaim - proclaim overlays)
NDI receive hardware acceleration	True
NDI alpha blending fix	False
Purpose	Visual Proclaim overlay/transparent livestream channel. This source is muted and hidden from the audio mixer in the saved scene collection.

6.4 Intro_Video media source

Setting	Current value / note
OBS source name	Intro_Video
OBS source type	ffmpeg_source
Enabled / muted	enabled=True; muted=True
Volume	linear=1.0 (0.0 dB)
Audio sync offset	0 ms
Monitoring	Monitor off
Mixer track note	OBS mixer bitmask=255. Track 1 is selected in the profile output, and these sources include Track 1.
Local media file	C:/ChurchAutomation/Intro Video/20_second_greeting.mp4
Hardware decode	True
Close file when inactive	True
Restart playback when source becomes active	False
Purpose	Greeting/intro video used in the Introduction scene and restarted by Stream Agent at stream start.

WARNING: The saved scene collection shows Intro_Video muted. If the intro video is expected to provide audio, verify this in OBS before Sunday. If the desired audio is only the live Axis camera/mixer feed during the intro, leaving the intro media muted is consistent with the Axis-embedded-audio design.

7. Audio design and mixer-track notes

The current design uses Axis embedded audio rather than a separate shared ASIO audio source. The West_axis and East_axis sources carry the camera video and the embedded audio together. Stream Agent controls OBS source volume targets for the Axis audio mode; it does not receive, process, or route the audio itself.

Item	Current state / implication
West_axis and East_axis	Not muted; volume about -2.6 dB; filters applied; embedded Axis audio is active.
slides	Muted and mixer-hidden; intended as visual Proclaim overlay only.
Intro_Video	Muted and mixer-hidden in the saved JSON. Verify if intro audio is desired.
OBS audio track used by stream/record	Track 1. Streaming TrackIndex=1 and recording RecTracks=1.
Audio sample rate	48 kHz. Keep Windows, camera audio, and any future mixer/interface path at 48 kHz where possible.
Audio sync offsets	All listed sources have OBS sync offset 0 ms in the scene collection.

Note: The old ASIO/USB audio path is not the active OBS audio path in this scene collection. It could be restored later by adding or re-enabling a shared audio input and changing Stream Agent AUDIO_MODE, but that would be a configuration change rather than the current documented setup.

8. OBS WebSocket and Stream Agent relationship

Setting	Current value / role
OBS WebSocket enabled	True
Port	4455
Authentication required	False
OBS WebSocket version seen in log	5.6.3
Stream Agent expectation	Stream Agent controls OBS through WebSocket. It switches scenes, starts/stops streaming, restarts the intro media, reads status, and adjusts configured OBS audio targets.

WARNING: OBS WebSocket authentication is currently off. That is convenient for Stream Agent on the private AV LAN, but the WebSocket port should not be exposed to the public internet. Remote access should be through trusted private-network methods only.

9. Rebuild and verification checklist

1. Install OBS Studio and required plugins: OBS WebSocket, DistroAV/NDI, OBS VST support, and the TDR Nova VST plug-in if the current EQ filters are to be preserved.
2. Restore or create the OBS profile named exactly NHLC.
3. Restore or create the scene collection named NHLC Axis Audio.
4. Set video to 1920 x 1080 base/output, 30 fps, Bicubic scaling, NV12, Rec.709, Partial range.

5. Set audio to 48 kHz, Stereo, default monitoring device unless a future site decision changes it.
6. Set Advanced Output streaming and recording to AMD hardware H.264 using the documented bitrate and 2-second keyframe interval.
7. Confirm recording path exists: C:\Stream Recordings.
8. Confirm service output uses custom RTMPS to YouTube and install the real production stream key only on the OBS PC.
9. Confirm scenes exist exactly: Introduction, West View, East View.
10. Confirm sources exist exactly: West_axis, East_axis, slides, Intro_Video.
11. Open West_axis and East_axis properties. Confirm RTSP TCP, hardware decode on, restart on activate off, reconnect delay 1 second, and audio=1 in the Axis URL.
12. Open the Audio Mixer. Confirm West_axis and East_axis are not muted. Confirm slides and Intro_Video are intentionally muted unless a different operating decision is made.
13. Start Stream Agent and confirm it connects to OBS WebSocket on port 4455.
14. Use Stream Agent Director / Web HUD to switch West View and East View and verify that each scene shows the correct camera and carries audio.
15. Perform a short private or test stream before relying on the restored setup for a Sunday service.

10. OBS-studio folder backup and global restore

This section documents the complete OBS configuration-folder backup that was captured for the current NHLC setup on the BeelinkOBS AVTECH PC. The current backup folder is C:\OBS_Backup_2026-06-27. The active OBS configuration folder on that PC is %APPDATA%\obs-studio. The complete backup can be used as a global restoration of OBS settings when a full recovery is needed.

Item	Current documented value / instruction
Current backup folder on BeelinkOBS	C:\OBS_Backup_2026-06-27
Current active OBS settings folder on BeelinkOBS	%APPDATA%\obs-studio
Typical expanded live path	C:\Users\<WindowsUser>\AppData\Roaming\obs-studio The exact user folder depends on the Windows account used on BeelinkOBS.
What a global restore does	Replaces the whole OBS configuration state: profiles, scenes, global settings, selected plugin settings, and related saved OBS data.
When to use it	Use for disaster recovery, replacement PC setup, or rollback after OBS settings are badly damaged. Do not use casually during a normal service day.

Security note: the complete OBS backup can contain sensitive information such as YouTube service settings, browser/plugin cache data, and camera RTSP URLs. Keep it private. Do not place the full backup ZIP or the full obs-studio folder in GitHub.

Recommended global restore procedure

Use this method only when you want OBS to return to the complete saved state from the backup folder.

1. Close OBS and Stream Agent III sd before restoring.
2. Confirm no OBS process remains in the system tray or Task Manager.
3. Locate the current BeelinkOBS backup folder: C:\OBS_Backup_2026-06-27.
4. Open the live OBS settings folder: %APPDATA%\obs-studio.
5. Preserve the existing live obs-studio folder by renaming it or copying it to a dated safety backup.
6. Copy the contents of C:\OBS_Backup_2026-06-27 into the live obs-studio folder.
7. Launch OBS and verify profile NHLC and scene collection NHLC Axis Audio.
8. Verify scenes Introduction, West View, and East View and sources West_axis, East_axis, slides, and Intro_Video.
9. Open Settings > Stream and install the real production stream key only on the protected OBS PC if needed.
10. Start Stream Agent, confirm OBS WebSocket connection on port 4455, and run a short private/test stream before relying on Sunday service.

Optional PowerShell restoration pattern

The following commands show the general idea. The backup path below reflects the current BeelinkOBS backup location. The commands rename the current live OBS folder first, then copy the saved backup into the live OBS location.

```
$backup = "C:\OBS_Backup_2026-06-27"
$live   = "$env:APPDATA\obs-studio"
$save   = "$env:APPDATA\obs-studio_before_restore_$(Get-Date -Format yyyyMMdd_HHmmss)"
Stop-Process -Name obs64,obs32 -ErrorAction SilentlyContinue
if (Test-Path $live) { Rename-Item $live $save }
Copy-Item $backup $live -Recurse
```

Important: the $backup line has been set for the current BeelinkOBS backup folder. If the backup is moved to a USB drive or another folder, edit the $backup line first.

Partial restore alternative

For small repairs, a partial restore is safer than replacing the whole OBS configuration folder. A partial restore normally means copying only the NHLC profile folder and the NHLC_Axis_Audio scene-collection file. This preserves other OBS data and reduces the chance of copying old plugin cache or unrelated profiles.

Partial restore item	Source under backup folder	Live OBS destination
NHLC profile	basic\profiles\NHLC	%APPDATA%\obs-studio\ basic\profiles\NHLC
NHLC Axis Audio scene collection	basic\scenes\NHLC_Axis_Audio.json	%APPDATA%\obs-studio\ basic\scenes\NHLC_Axis_Audio.json
OBS WebSocket config, only if needed	plugin_config\obs-websocket\config.json	%APPDATA%\obs-studio\ plugin_config\obs-websocket\config.json

11. Known cautions and items to verify later

The following items should be checked when the setup is rebuilt, restored from backup, or changed. They are not necessarily faults; they are the main places where a small mismatch can affect Sunday operation.

Caution / item	Why it matters	Recommended check
Real YouTube stream key must stay private	The documentation uses a redacted placeholder. A real stream key is equivalent to a password.	Store the real key only in OBS on the protected streaming PC or in a private secured backup. Do not commit it to GitHub.
Full OBS backup may contain sensitive plugin/browser data	A complete obs-studio folder can include cached browser/plugin state and service metadata.	Keep the full backup private. Use the documentation PDF/DOCX, not the raw backup, for public sharing.
Camera RTSP URLs can contain credentials	The saved OBS scene file may include camera login information inside source URLs.	Documentation should keep URLs redacted as rtsp://<camera-login>@... unless the private recovery file is being used by a trusted maintainer.
Intro_Video is muted in the saved scene collection	If intro music or greeting-audio is desired, the OBS mute state must match that operating decision.	Before Sunday, play the intro and confirm whether muted intro video is still the desired design.

Current transition in backup is Fade at 350 ms	Stream Agent can force Cut or allow OBS transitions depending on its configuration.	Confirm whether the service plan is Cut or Fade and keep Stream Agent and OBS aligned.
Recording format is MP4	MP4 is convenient but can be less crash-safe than MKV if the PC loses power or OBS crashes.	Keep MP4 if operational convenience is preferred. Consider MKV plus auto-remux only after testing.
OBS WebSocket authentication is off	This is convenient on the private AV LAN but unsafe if exposed to the internet.	Do not expose port 4455 publicly. Use trusted private-network remote access only.
DistroAV/NDI and TDR Nova must be installed	The scene collection expects the Proclaim NDI source and VST filter chain to exist.	On a rebuilt PC, install and verify these before judging the OBS scene collection as complete.
C:\Stream Recordings must exist	OBS recording can fail if the configured recording folder is missing or inaccessible.	Create the folder and perform a short test recording after restore.

Screenshots to add in a later revision

- OBS Settings > Output, showing streaming and recording encoder settings.
- OBS Settings > Video, showing 1920 x 1080 and 30 fps.
- OBS Settings > Stream, with the stream key hidden or redacted.
- OBS Audio Mixer, showing West_axis and East_axis active and slides/Intro_Video muted as intended.
- Properties windows for West_axis, East_axis, slides, and Intro_Video.
- OBS WebSocket settings window showing port 4455 and the local private-network assumption.